# Jean-Baptiste Saulnier *Quantitative Biologist*

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ORCID: 0009-0009-3099-2307 French citizen

## RESEARCH EXPERIENCE

2025 – present **Postdoctoral Fellow**
*Institute of Plant and Microbial Biology, Academia Sinica, Taipei, Taiwan*
Investigating membrane potential regulation in *Bacillus subtilis* biofilms at both single-cell and community scales. Combining mathematical modeling, custom microfluidic chip fabrication, and multiscale microscopy. Supervisor: Dr. Jian-geng Chiou.

2024 – 2025 **CNRS Postdoctoral Fellow**
*Laboratoire de Chimie Bactérienne (LCB), Marseille, France*
Studied topological defects in collective patterns of *Myxococcus xanthus* mutants to characterize organizational phenotypes. Supervisor: Dr. Tâm Mignot.

2019 – 2023 **PhD in Biophysics**
*Laboratoire de Chimie Bactérienne (LCB), Marseille, France*
Elucidated the mechanism of spatial pattern transition between rippling and swarming in *Myxococcus xanthus* during predation, combining mathematical modeling, image analysis, and molecular biology. Supervisors: Dr. Tâm Mignot & Dr. Vincent Calvez.

## PUBLICATIONS

2024 **The mechanism of spatial pattern transition in motile bacterial collectives**
**J.-B. Saulnier**, M. Romanos, J. Schrohe, C. Cuzin, V. Calvez, T. Mignot
*Preprint* — bioRxiv 10.1101/2024.10.28.620572

2023 **A new modeling approach of *Myxococcus xanthus* bacteria using polarity-based reversals**
H. Bloch, V. Calvez, B. Gaudeul, L. Gouarin, A. Lefebvre-Lepot, T. Mignot, M. Romanos, **J.-B. Saulnier**
*CEMRACS Proceedings* — HAL hal-04102694

## EDUCATION

2018 – 2019 **Master's in Theoretical Physics**
Aix-Marseille Université. Developed a reversal-detection algorithm for *M. xanthus* using tracking and image analysis. Supervisor: Dr. Tâm Mignot.

2016 – 2018 **Master's in Cosmos, Fields and Particles** — *Université de Montpellier*

2013 – 2016 **Bachelor's in Fundamental Physics** — *Université de Montpellier*

2011 – 2013 **DUT in Thermal Engineering and Energy** — *Université de Belfort-Montbéliard*

## SUPERVISION, CONFERENCES & TRAINING

*Supervision*

2023 (6 mo.) Undergraduate student supervision (Jonathan Schrohe) — *SgmX polar protein tracking*

*Conferences & Training*

2025, 2026 Q-Microbio conference — *Quantitative Microbiology*

2024 Frontiers in Interacting Particle Systems — *1 wk* · Mathematical Biology: Collective Behavior and Pattern Formation — *1 wk, CIRM, Marseille*

2022 CENTURI Hackathon: Hacking Living Systems — *CIRM, Marseille*

2022 EMBO Workshop: Adherent Microbial Communities — *1 wk* · **CEMRACS** — *5 wks*

2021 Fall School in Complex Systems — *1 wk, Nice · 1 wk, Fréjus*

## SKILLS

Modeling: PDE, agent-based simulations · Microscopy: phase contrast, epifluorescence · Microfluidics: wafer to chip (photolithography, PDMS) · Programming: Python, SageMath · Languages: English (fluent), French (native)